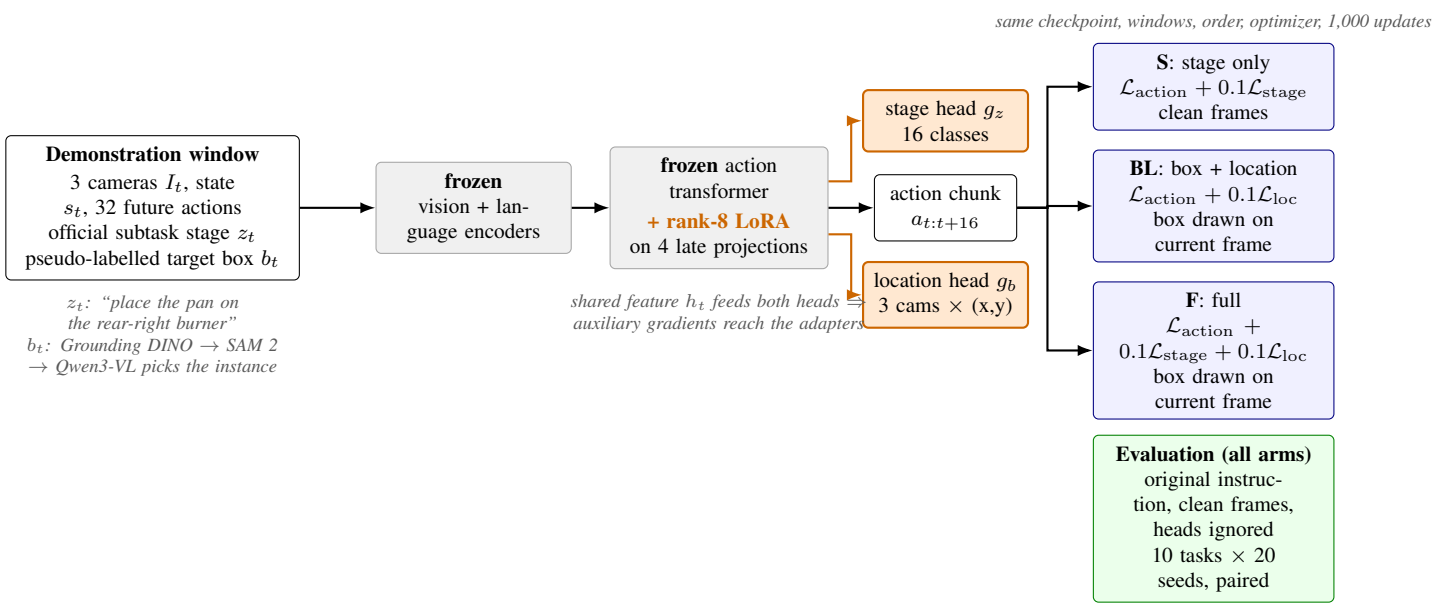

same checkpoint, windows, order, optimizer, 1,000 updates
Demonstration window
3 cameras $I_t$, state
$s_t$, 32 future actions
official subtask stage $z_t$
pseudo-labelled target box $b_t$
$z_t$: "place the pan on the rear-right burner"
$b_t$: Grounding DINO → SAM 2 → Qwen3-VL picks the instance
frozen
vision + language encoders
frozen action transformer
+ rank-8 LoRA
on 4 late projections
shared feature $h_t$ feeds both heads
auxiliary gradients reach the adapters
stage head $g_z$
16 classes
action chunk
$a_{t:t+16}$
location head $g_b$
3 cams × (x,y)
S: stage only
$\mathcal{L}_{\text{action}} + 0.1\mathcal{L}_{\text{stage}}$
clean frames
BL: box + location
$\mathcal{L}_{\text{action}} + 0.1\mathcal{L}_{\text{loc}}$
box drawn on current frame
F: full
$\mathcal{L}_{\text{action}} + 0.1\mathcal{L}_{\text{stage}} + 0.1\mathcal{L}_{\text{loc}}$
box drawn on current frame
Evaluation (all arms)
original instruction, clean frames, heads ignored
10 tasks × 20 seeds, paired